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## SAVE: saliency-assisted volume exploration

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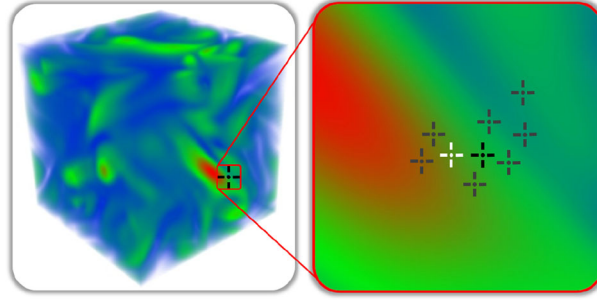
**Abstract** Interactive visualization has become a valuable tool in visual exploration of scientific data. One prerequisite and fundamental issue is how to infer three-dimensional information through users' two-dimensional input. Existing approaches commonly build on the hypothesis that user input is precise, which is sometimes invalid because of multiple causes like data noise, limited resolution of display devices and users' casual input. In this paper, we reconsider some design choices of previous methods and propose an alternative effective algorithm for inferring interaction position in scientific data, especially volume data exploration. Our method automatically assists user interaction with the defined saliency. The presented saliency integrates data value, corresponding transfer function and user input. The result saliency implies remarkable regions of raw data as existing methods. Moreover, it reflects the areas of users' concern. Thirdly, it eliminates the errors from data and device, helping users get the region they focus on. Various experiments have verified that our method can reasonably refine user interaction and effectively help users access interested features.

**Keywords** Interactive visualization · WYSIWYG · Human perception

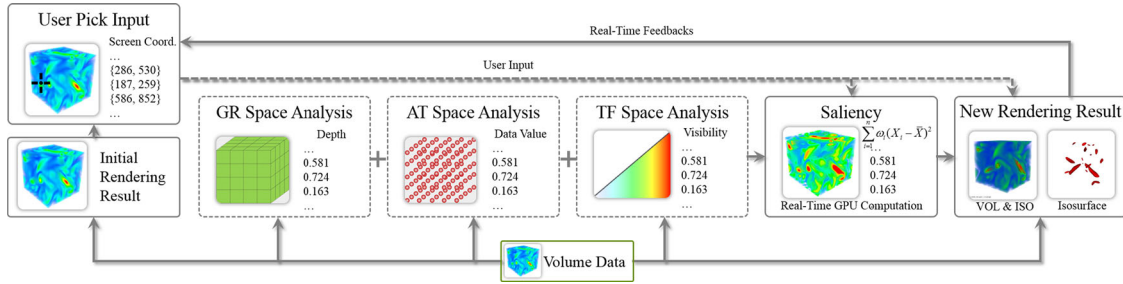
### 1 Introduction

Interactive visualization (IV) has been widely used in various areas, especially where certain data features are difficult to describe mathematically or to anticipate prior to the analysis (Kehrer and Hauser 2013). And it is always regarded as a challenge in visualization community (Hibbard 1999; Johnson 2004; Laramée and Kosara 2007). Generally, IV has to cope with a prerequisite problem, which is how to infer interaction position in three dimension (3D) based on users' two-dimensional input, namely *Where To Pick* (WTP). It is necessary in Scientific Visualization, where at the user always explore three-dimensional data with two-dimensional input devices (like mouse).

The WTP problem has been studied for a long time (Brodbeck et al. 2009). To our knowledge, existing approaches commonly build on the hypothesis that users' interaction is precise and do not take human perception into account. In practice, users' interaction is inclined to make errors due to multiple factors, including objective ones (such as limited data resolution, data noise, finite display device resolution, and interaction device errors) and subjective ones (such as retina resolution, fat-finger problems, and customary interaction with certain degree of arbitrary). Moreover, it is believed that sensory perceptual system of human depends on the relative judgment about things (Weber's Law), including color, luminance and size (Shen 2003), all of which are used frequently in visualization. It is found that this property of human could



**Fig. 1** WTP illustration without loss of generality, if users click on the position shown with *black cross* on the *left*, a number of interactive positions are possible in detail, as shown with crosses on the *right*. The causes are manifold as discussed in Sect. 1. Yet at present, strictly corresponding methods are mostly taken and thus corresponding positions are got as shown with *black cross*, which are sometimes error-prone as illustrated in Sect. 5. The position shown with *white cross* is the revised position with our method



**Fig. 2** The pipeline of our framework. The user input is revised with the proposed saliency, which is computed with Grid (GR), Attribute (AT), Transfer Function (TF) space and user input. Result volume (VOL) and/or isosurface (ISO) gives users real-time feedback with precise interaction information. Please refer to Sects. 3 and 4 for overviews and details respectively

lead to misjudgment as well. As shown in Fig. 1, without loss of generality, if users click on a position as shown with black cross on the left, there are some possible candidate positions as shown with crosses on the right. At present, strictly corresponding methods are mostly taken and thus corresponding positions are got, as shown with black cross on the right.

Inspired by recent research progresses in human perception (Goldstein 2013) and human computer interaction (Dix 2009), we reconsider some design choices of previous methods and propose an alternative effective algorithm to cope with WTP problems in scientific data, especially volume data exploration, which applies volume saliency to assisting visualization interaction, called Saliency-Assisted Volume Exploration (SAVE) as illustrated in Fig. 2. Our method is based on the observation that users explore their data visually trial and error. Our method tries to revise user's interaction input automatically with the defined saliency, providing a **WYSIWYF (What You See Is What You Focus)** schema. To demonstrate the utility of this method, we use it to get the revised input position in 3D, and then get the corresponding data value to extract isosurface.

The main contributions of this paper are briefly as follows:

- We propose a definition of volume saliency integrating grid, attribute, opacity and user input;
- We present an effective and convenient algorithm to revise user interaction based on defined saliency;
- We test our new method with practical datasets from various areas and demonstrate that our method is widely applicable.

## 2 Related work

In this section, we review the previous works on saliency in visualization and picking in volume visualization. This work is mainly concerned with improving interaction for visualization based on perception principles, rather than analyzing how transfer function is designed or how features are extracted.

## 2.1 Saliency

The saliency of an item, such as pixel, is the state or quality by which it stands out relative to its neighbors. Saliency detection is considered a key attentional mechanism that facilitates learning and survival by enabling organisms to focus their limited perceptual and cognitive resources on the most pertinent subset of the available sensory data (Zhao and Koch 2013). Saliency detection has been studied in computer vision for years (Koch and Ullman 1987; Koch and Poggio 1999). Recently, saliency is applied to computer graphics (Lee et al. 2005) and visualization (Kim and Varshney 2006; Yu et al. 2012).

Kim and Varshney (2006) introduced a saliency-based enhancement operator to guide visual attention in volume visualization and augmented the existing visualization pipeline by incorporating enhancement operators to increase the visual saliency of different regions of a volume dataset. However, they compute saliency field with data value only. Their method deals with the problem that highlights features near user interaction with spherical regions of interest, while the method in this paper handles the problem inferring picking position with the defined saliency.

Yu et al. (2012) proposed flow saliency to assist streamline seed selection. The generated set of streamlines captures important flow features near critical points without enforcing the dense seeding condition. The defined flow saliency integrates curvature and torsion based on the Gaussian weighted center-surrounding (Itti et al. 1998) evaluation. This method only takes geometry of streamline into consideration, ignoring other information, like data attributes. As a result, it could only capture region with obvious shape-changing streamlines.

Saliency could be also used to evaluate the quality of visualization (Janicke and Chen 2010), performing progress visualization (Machiraju et al. 2001). Generally, a saliency map is computed from the data using center-surround operation and Gaussian-weighted average of the emphasis field centered at a voxel (Itti et al. 1998). Our method also uses Gaussian-weighted center-surround operation in volume data attribute space.

## 2.2 Picking

Interactive visualization needs to cope with the problem that where users really want to interact or pick in space. Researchers have made many effects on this problem.

Gobbetti et al. (1998) introduced the most widely used volume picking method, which searches along the ray that passes the focus point and user input position in screen, and then comes through the volume data. The search stops as soon as the accumulated opacity exceeds a user-defined threshold. The endpoint of the search is returned as a 3D position. The threshold could be zero (first-hit), user defined opacity or data value threshold (Toennies and Derz 1997).

Kohlmann et al. (2009) presented contextual picking for volumetric structures, especially tailored to medical data in DICOM images. They firstly extract meta information to automatically select contextual profiles from the knowledge base. Then, the current ray profile is analyzed to detect anatomical structures after users pick, which are represented by the selected contextual profiles.

Wiebel et al. (2012) proposed a What You See Is What You Pick (WYSIWYP) technique. The technique gets picking positions by computing the largest “Jump” interval of the accumulated opacity along the ray. As related work, this method relies on the user input and builds upon the assumption that the input position in the screen is precise. This assumption causes these methods to be susceptible to error or noise and inflexible at times (see Sect. 1). Some typical cases were also discussed in Wiebel et al. (2012).

## 3 Overview

In this section, we present an overview of our method based on the defined computational models of saliency as shown in Fig. 2.

- *Brief pipeline* the pipeline of the schema (Fig. 2) starts from the initial rendering result and the user input. And then the user input is revised based on the computed local saliency. Our method gives real-time image feedback, which allows users to progressively explore their volume.
- *Saliency field* we assume that a saliency value is assigned to each voxel of the volume data as exiting methods. But our saliency assignment integrates voxel position, attribute value, opacity and user interaction. The saliency field is updated in real time while user interacts with rendering results.

- *User input* users interact with the volume rendering results directly on the screen. The interaction input is simply 2D screen position. Meanwhile, the interaction is assisted with precise feedback including position in screen and space coordinate, picked data value and corresponding color (Shen et al. 2012).

## 4 SAVE

### 4.1 Saliency

From human perception, positions, attributes and opacity are important properties of voxels in volume, as these three quantities capture the intrinsic nature of each individual voxel in grid space, attribute space and transfer function space. User input indicates interested feature in specific regions. It is necessary to consider user's input in saliency computation to reflect users' concern.

Let a voxel  $v_i$  be the  $i$ -th voxel in a volume  $V$ . Then, let  $a_i$  be the attribute value for a voxel  $v_i$ . It is known that the gradient  $grad(a_i)$  points in the direction of the greatest rate of increase of the attribute and its magnitude is the slope of the attribute in that direction. It is further confirmed with practical data as shown in Fig. 3, which displays the whole attribute data, gradient field and second-order gradient field from left to right. To observe clearly, parts of the fields in Fig. 3 are shown in the top left Fig. 4a–c. And the bottom left Fig. 4d–f is the magnitude curve of the corresponding fields along the diagonal line. The right Fig. 4g compares these three fields after normalization, through which it is found that the peaks of gradient are corresponding to the range of raw data value with maximum variation in the local region. Moreover, it is well known that human's vision system is more sensitive to variation. The gradient is used to compute saliency. It is necessary to notice that only gradient which does not integrate local relations is not enough to capture salience regions.

The saliency is based on the Gaussian weighted center-surrounding evaluation Itti et al. (1998). Let  $S_{a_i}$  be the attribute saliency value for a voxel  $v_i$ .  $\mathcal{G}(|grad(a_i)|, \sigma_i)$  denotes the Gaussian-weighted average of the gradient magnitude of the attribute at voxel  $v_i$ , where  $\sigma$  is the threshold distance which belongs to neighboring points of voxel  $v_i$ . The attribute saliency is defined for a voxel  $v_i$  using the Gaussian weighted averages computed at fine and coarse scales as (Lee et al. 2005):

$$S_{a_i} = |\mathcal{G}(|grad(a_i)|, \sigma_i) - \mathcal{G}(|grad(a_i)|, 2\sigma_i)| \quad (1)$$

where  $\sigma_i$  is the standard deviation of the Gaussian filter at scale  $i$ . Similarly, the opacity saliency is defined as,

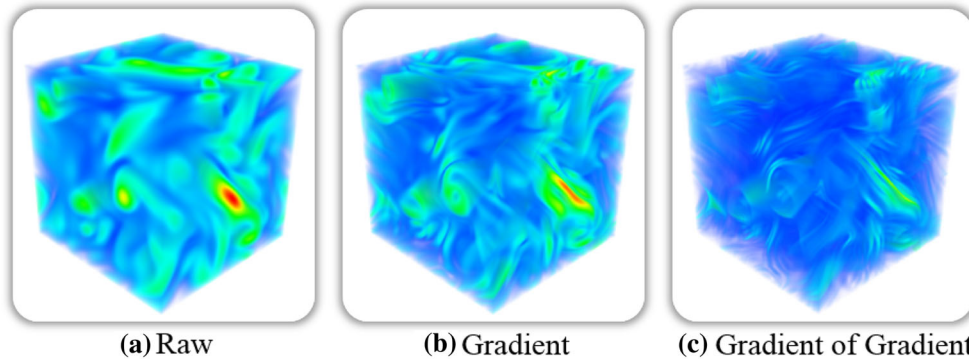
$$S_{o_i} = |\mathcal{G}(|grad(o_i)|, \sigma_i) - \mathcal{G}(|grad(o_i)|, 2\sigma_i)| \quad (2)$$

where  $o_i$  is the opacity value of voxel  $v_i$ . In practice,  $\sigma_i = 4\epsilon$ , where  $\epsilon$  is the grid size of the data.

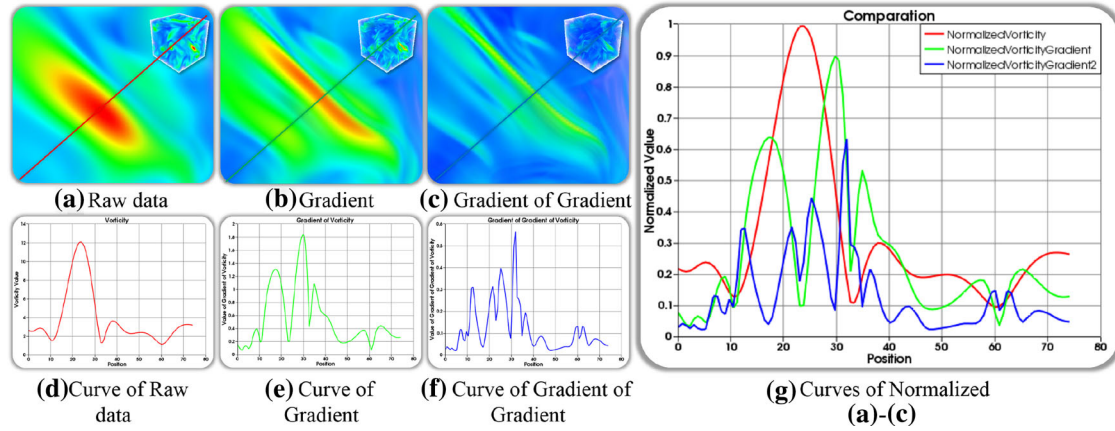
It is valuable to notice that the attribute saliency could be extended to cope with multi-variant data sets,

$$S_{a_{ij}} = |\mathcal{G}(|grad(a_{ij})|, \sigma_{ij}) - \mathcal{G}(|grad(a_{ij})|, 2\sigma_{ij})| \quad (3)$$

where  $a_{ij}$  stands for the  $j$ -th attribute of multi-variant data sets at voxel  $v_i$ .



**Fig. 3** The validation of attribute gradient for saliency computation. For clarity and comparison, the raw attribute field (a), gradient field (b) and second-order gradient field (c) are visualized from left to right. Blue and red colors are corresponding to small and large field value, respectively



**Fig. 4** Gradient observation in detail. Parts of the three fields in Fig. 3 are shown from **a** to **c**. And **d–f** is the *magnitude curve* of the corresponding fields along the *diagonal line*. Red, green, and blue curves are corresponding to **a–c**, respectively. **f** compares these three fields in **a–c** after normalization

Let  $N(f)$  be the normalization of function  $f$ . So,

$$N(S_{a_i}^m) = \sum_{j=1}^k \omega_j (N(S_{a_{ij}})) \quad (4)$$

where  $S_{a_i}^m$  is the overall attribute saliency at voxel  $v_i$ ,  $\omega_j$  is the weight of  $N(S_{a_{ij}})$  in the computation of  $N(S_{a_i}^m)$ ,  $k$  is the number of variants.

To emphasize interested local features, we define a distance factor based on the user input. Let  $p(x, y)$  be the position user clicked on the screen, and  $d'_{pi}(x, y)$  be the pixel distance between voxel  $v_i$  and  $p(x, y)$  on screen coordinate. The distance factor is defined as:

$$d_{pi}(x, y) = e^{-k \cdot N(d'_{pi}(x, y))} \quad (5)$$

where

$$k = \begin{cases} -1 & \text{if } d'_{pi}(x, y) \leq 5 \\ 1 & \text{if } d'_{pi}(x, y) > 5 \end{cases} \quad (6)$$

Then we get the result saliency:

$$S_{v_i} = (N(S_{a_i}) + N(S_{o_i})) \cdot d_{pi}(x, y) \quad (7)$$

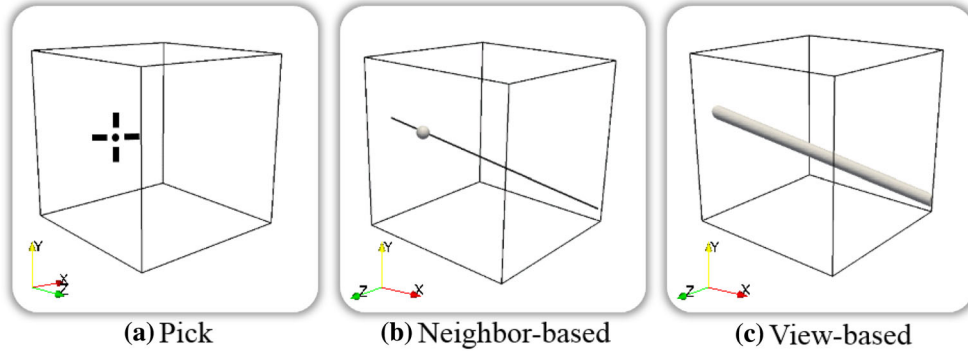
Finally, the interaction position in space is the place with the local max saliency.

## 4.2 Implementation details

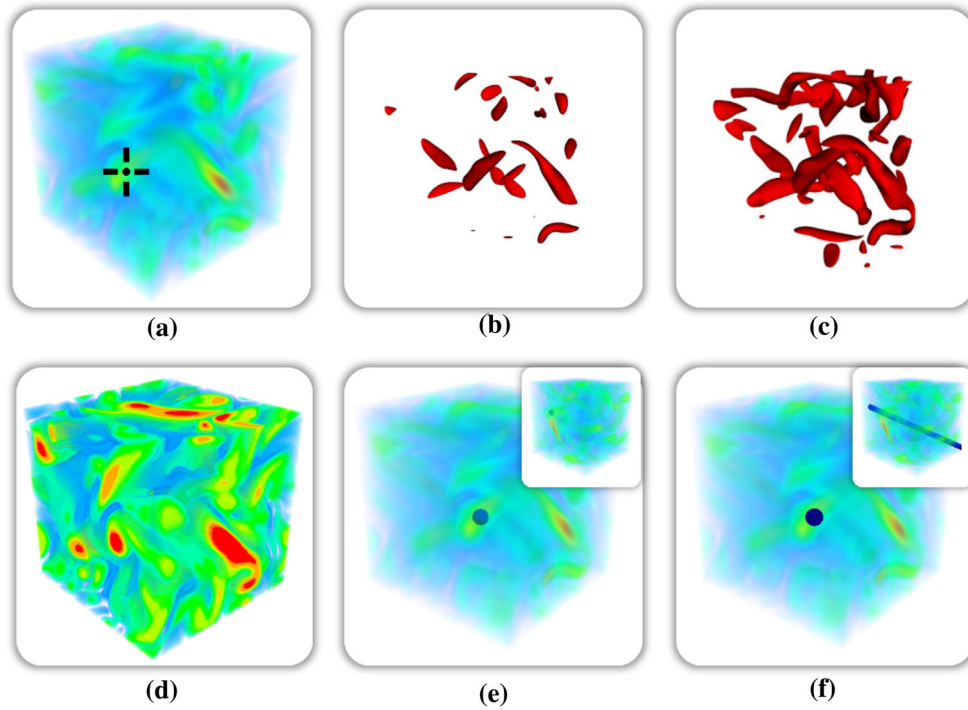
In practice, after the data are read, the attribute saliency is computed. The opacity saliency needs to be computed only when the transfer function is changed. In most situation, we only need to recompute the distance factor and the result saliency field when users interact with the data sets. With CUDA, we got the saliency field in real time in cases as shown in Sect. 5.

For comparison, we do not only implement the pixel distance in computing the distance factor as shown in Fig. 5c, but also realize the Euclidean distance computation in which the space position of picking is computed by the State-of-The-ART (STAR) WYSIWYP method (Wiebel et al. 2012). As users click on the volume as shown with black cross in Fig. 5a–c is the results from other views of observation. The neighbor-based mode (as shown in Fig. 5b) firstly computes space position using the WYSIWYP method, then the distance factor in the gray sphere is computed with  $k = -1$ , while other region is set with  $k = 1$  in Eq. (5). The view-based mode (as shown in Fig. 5c) is corresponding to the pixel distance, which sets  $k = -1$  in the gray cylinder region, while other region is set with  $k = 1$  in Eq. (5). Figure 6 shows the comparison directly. It illustrates that the view-based mode could get vortex with larger scale than the result got by neighbor-





**Fig. 5** Different distance modes in computing the distance factor. *Left* users click on the volume as shown with *black cross*. *Middle* and *Right* is the result from other views of observation. *Middle* the neighbor-based mode. *Right* the view-based mode



**Fig. 6** Different distance mode comparisons: **a** pick on the raw volume data; **b** result computed with neighbor-based mode; **c** result computed with view-based mode; **d** the computed saliency field with view-based mode. **e–f** is the illustration of Fig. 5b, c. We can clearly get that the view-based mode could get vortex with larger scale than the result got by neighbor-based mode

based mode. Discussion with Computational Fluid Dynamics (CFD) experts has confirmed that Fig. 6c is practically what they want. More experiments have proved that the pixel distance mode is more effective than the neighbor-based mode.

## 5 Results

In this section, we first introduce the datasets and timing performance, and then the experiment results are discussed in detail.

## 5.1 Data sets and timing performance

We experimented our method with six volume data sets. The Turbulent data set is from a turbulent simulation provided by Dr. D. Silver at Rutgers University. The delta-wing data set is a simulation of flow past a geometry representing a delta wing aircraft. The particle data set is from a simulation of the spreading of the energy in a high-speed simulation. H Atom data set is from the SFB 382 of the German Research Council (DFG). The CT Head data set is from the Visual Human Project. Finally, the combustion data set is from a turbulent combustion simulation performed at the Sandia National Laboratories. Table 1 lists these data sets and the timing for every step of our algorithm.

The performance of the proposed method is tested on a desktop computer, with a quad-core Intel i7-950 2.8 GHz processor with 12 GB memory, and an Nvidia Geforce K5000 graphics card with 4096 MB video memory. The timings of the data sets are listed in Table 1. The computation of attribute saliency, opacity saliency, distance factor and saliency is performed using CUDA 6.0. The timing mainly reflects the complexity of each step.

## 5.2 Results

In this section, we discuss the test and verification of the proposed method in detail. It is necessary to reemphasize that the result isosurface is used to visually display the data value at the revised interaction position in space and it is helpful for compare with present methods. Please refer to our attached video for vivid illustrations of all these tests.

### 5.2.1 Turbulent vortex

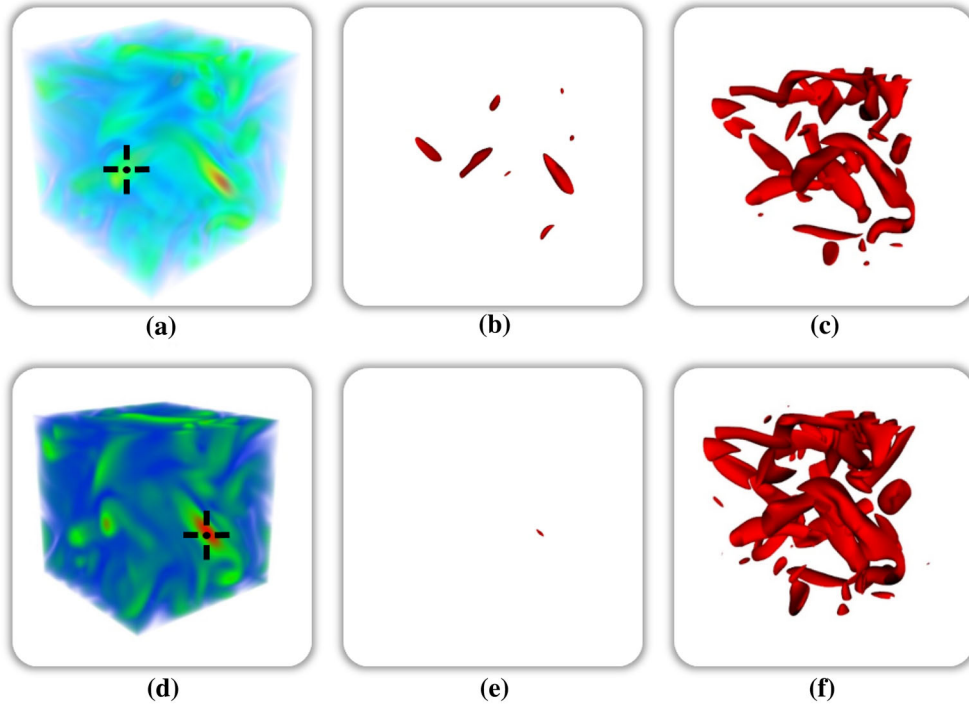
Turbulent Vortex data set consists of vorticity value in the turbulent field. The value describes the magnitude of the local spinning motion of a fluid near some point. Experts from CFD state that for complex vortex data set, like turbulent vortex, they always observe the distribution of some interested value which is typically predicated with experience or retried with the help of slice and/or volume rendering. The prior way is depended heavily on experience and difficult to deal with new data sets. The latter method is error and trial, which is time-consuming. Therefore, we apply our method to the data set for testing its efficiency in getting focused features when users interact, as shown in Fig. 7. As users picked on the raw volume shown with black cross (Fig. 7a), our method (Fig. 7c) gets vortex with larger scalar than the WYSIWYP method (Wiebel et al. 2012) (Fig. 7b). Our results clearly reveal the possible boundary of certain vortex. Figure 7(d–f) tests and verifies our method with different transfer functions and picked positions. It is obvious that the WYSIWYP method (Wiebel et al. 2012) only gets part of vortex in smaller scalar, while our method could get the certified results as Fig. 7c. It is necessary to notice that in Fig. 7a user picked an internal feature with smaller opacity transfer function, while in Fig. 7d users pick a marginal feature with larger opacity transfer function. These cases have testified the effectiveness of our method with different transfer functions.

### 5.2.2 Delta wing

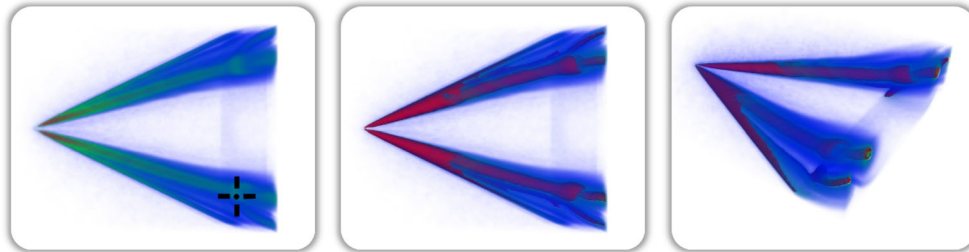
Delta Wing data set represents a simulation where flow pasts a very simplified geometry representing a delta wing aircraft at 40° angles of attack. Features of interest are vortices and vortex breakdowns (Ekaterinaris and Schiff 1990). First, the raw volume data are shown as Fig. 8a. Then if users are interested in the features of interest, they just click on the related region, such as the black cross marked in Fig. 8a. Then our method conveniently gives real-time feedback as shown in Fig. 8b, on which user could observe key features or

**Table 1** The data sets tested in our experiments, their configurations and timing performance in seconds

| Data set             | Dimension   | Attribute saliency (s) | Opacity saliency (s) | Distance factor (s) | Saliency (s) |
|----------------------|-------------|------------------------|----------------------|---------------------|--------------|
| Turbulent (Fig. 7)   | 128×128×128 | 0.223                  | 0.280                | 0.0016              | 0.0053       |
| Delta wing (Fig. 8)  | 256×256×128 | 0.509                  | 0.511                | 0.0030              | 0.0029       |
| Particle (Fig. 9)    | 512×512×620 | 14.027                 | 13.634               | 0.1032              | 0.0946       |
| H atom (Fig. 10)     | 128×128×128 | 0.223                  | 0.281                | 0.0016              | 0.0053       |
| CT head (Fig. 11)    | 256×256×159 | 0.547                  | 0.553                | 0.0034              | 0.0032       |
| Combustion (Fig. 12) | 256×256×256 | 1.084                  | 1.103                | 0.0070              | 0.0062       |



**Fig. 7** Turbulent Vortex: **a** user picks on the raw volume data as shown with *black cross*; **b** the result got by the WYSIWYP method (Wiebel et al. 2012); **c** the result got with our method; **d** user picks on the same raw volume data with **a** but different transfer functions and picked positions; **e** the result got by the WYSIWYP method (Wiebel et al. 2012); **f** the result got by our method



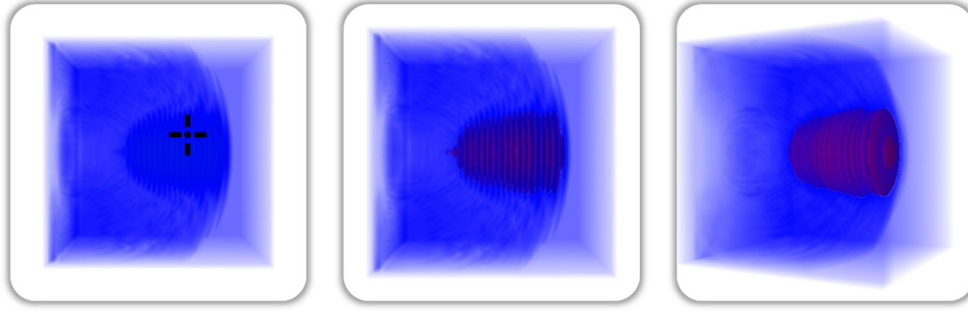
**Fig. 8** Delta Wing flow pasts a very simplified geometry representing a delta wing aircraft at  $40^\circ$  angle of attack. **a** the raw volume data and users pick position illustration. **b–c** the result with our method with different *angles* of view. Users could get features of interest, including vortices and vortex breakdowns (Ekaterinaris and Schiff 1990), with only one flexible click operation

make further exploration, like simply modification of angles of view in Fig. 8c. Based on SAVE, user could get features of interest, like vortices and vortex breakdowns, through only one flexible click operation.

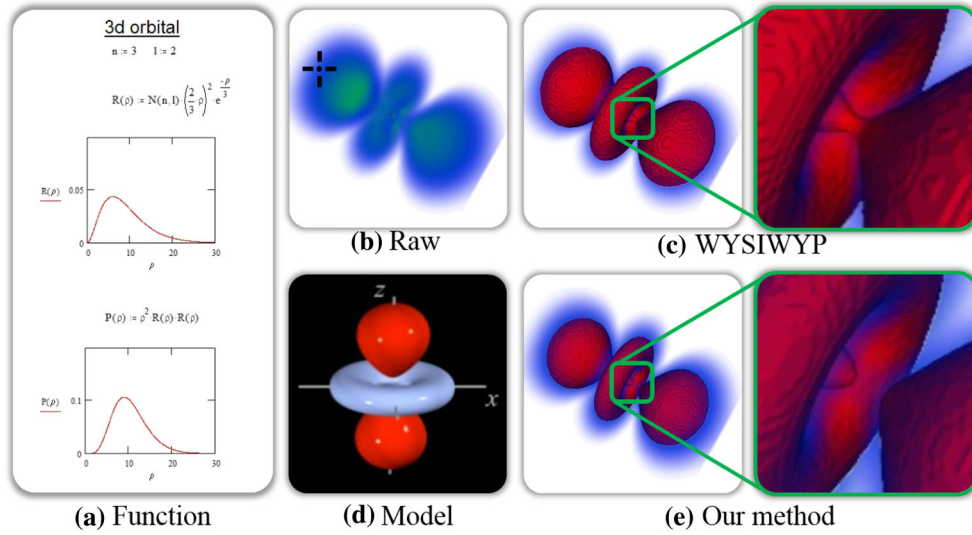
### 5.2.3 Particle

Particle data set represents the electromagnetic field generated by a high-speed particle-in-cell (PIC) simulation, which bases on the time-domain finite difference. Finite-difference time-domain (FDTD) algorithm and the concept of finite-size particle could solve the Maxwell equations and the motion equations of electrons directly (Luan et al. 2013). The physical model and computational technology are in cuboid mesh as shown in Fig. 9. With users picking on the central critical region like the position shown with *black cross* in Fig. 9a, the boundary of high electromagnetic energy is enhanced in the visualization result with





**Fig. 9** Particle high-speed particle-in-cell simulation. **a** the raw volume visualization of electromagnetic field and interaction position (black cross). **b–c** the result with our method with different angle of view. User could get clear comprehension of important edge of the high electromagnetic energy region with simple interaction



**Fig. 10** Hydrogen atom **a** probability of the likelihood of finding the electron in the cell's space (Nexus 2014); **b** users pick on the raw volume data as shown with black cross; **c** the results got by the WYSIWYP method (Wiebel et al. 2012); **d** the theoretical model of 3d orbital Hydrogen Atom (Nexus 2014); **e** the results got with our method. As green rectangles highlight, it can be found that only our results are consistent with the probability curve **a** and theoretical model **d**

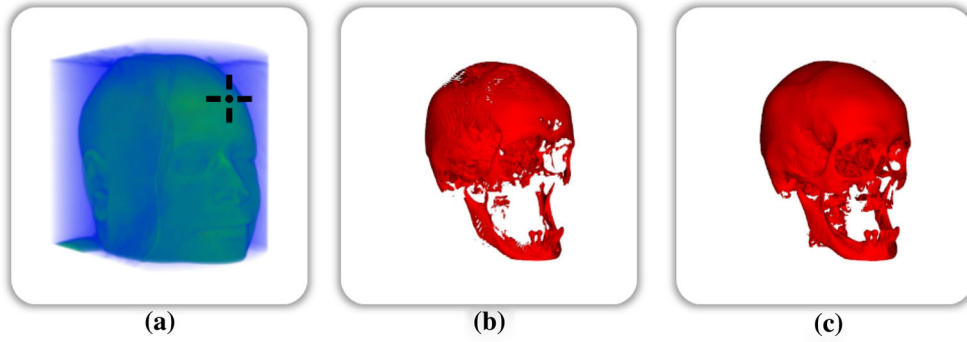
isosurface (Fig. 9b). Then users could explore the result further as shown in Fig. 9c. This case and prior one have verified that the SAVE method is helpful in picking interested features effectively.

#### 5.2.4 Hydrogen atom

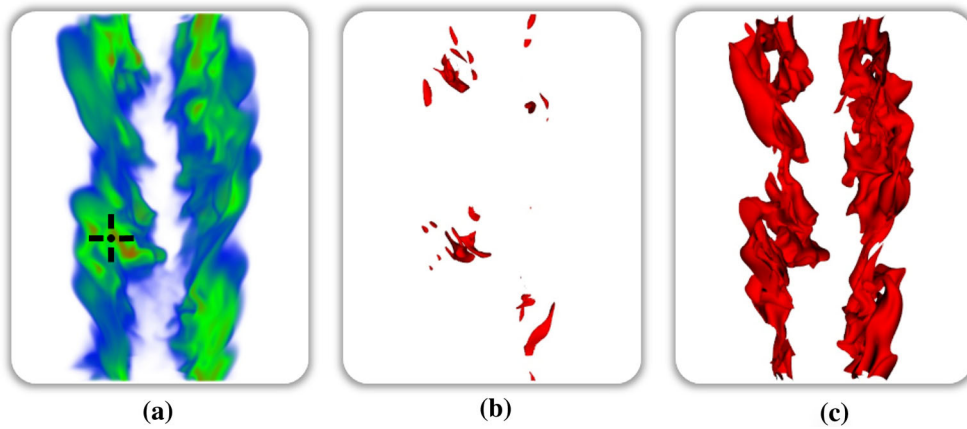
Hydrogen Atom data set represents the probability density of the electron of a hydrogen atom, residing in a strong magnetic field. The volume data are a 3D array, with each cell in the array representing a small volume of space located at that cell's  $(x, y, z)$  location. The value in each cell represents the likelihood of finding the electron in the cell's space at a given instant in time. Zero represents no probability, and one represents complete probability as shown in Fig. 10a (Nexus 2014), from which we could get that the probability of the electron on the center is zero. When users pick on the position as shown with black cross in Fig. 10b which was discussed as a failure case in Wiebel et al. (2012), it can be found that only our results reflect the probability correctly compared with the probability curve (Fig. 10a) and theoretical model (Fig. 10d).

#### 5.2.5 CT head

CT Head data set represents the structure of human head including several parts, like skeletons and faces. Present methods could fail to pick out skeletons in cases, where users pick on the position as shown with



**Fig. 11** *CT head* **a** users pick on the raw volume data as shown with *black cross*; **b** the result got by the WYSIWYP method (Wiebel et al. 2012); **c** the result got with our method. It is clear that the picked structure with our method is more complete than the WYSIWYP method (Wiebel et al. 2012)



**Fig. 12** *Combustion* represents hydroxyl radical ( $OH$ ) in the turbulent Combustion Simulation. **a** Initial volume rendering and user's picking position. **b** The result got by the WYSIWYP method (Wiebel et al. 2012); **c** the result got with our method. With the result from our method, users could observe continuous convoluted isosurface clearly, which is the surface of the mixture fraction and helpful in analysis of the combustion behavior

black cross in Fig. 11a. Compared with the result from the WYSIWYP method (Wiebel et al. 2012) (Fig. 11b) and the proposed method (Fig. 11c), the extracted skeleton through our method is more complete.

### 5.2.6 Combustion

Combustion data set represents hydroxyl radical ( $OH$ ) in the turbulent Combustion Simulation. The distribution is shown in Fig. 12a. If users pick on the position as shown with black cross, the results from the WYSIWYP method and our method are shown in Fig. 12b and c, respectively. With the result from our method, users could observe continuous convoluted isosurface clearly. The latter result reflects the surface of the mixture fraction and is helpful in combustion behavior analysis.

## 6 Conclusions and future work

Effective interaction with visualization of scientific data remains a challenge, which is getting exacerbated with the increase in the resolution of data sets ever produced. The saliency-assisted volume exploration method copes with the prerequisite WTP problem and offers an alternative way to effectively interact with the volume structure and patterns. Saliency-assisted method implicitly provides a WYSIWYP schema to help users pick focused features conveniently. The proposed saliency could be easily integrated with different low-dimensional or high-dimensional transfer functions.

The volume saliency definition can be modified to cope with 3D flow field or multi-variant datasets. In the future, we will improve the exploration process by updating transfer function based on volume saliency. Finally, although informal discussions have confirmed the effectiveness of the method, it is valuable to make a systematic formal evaluation of the method and involve domain scientists in the evaluation of this new approach.

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